AQUARIS
MADE IN ITALY



METERING PUMP

JLM · JBB SERIES

SELECTION BROCHURE

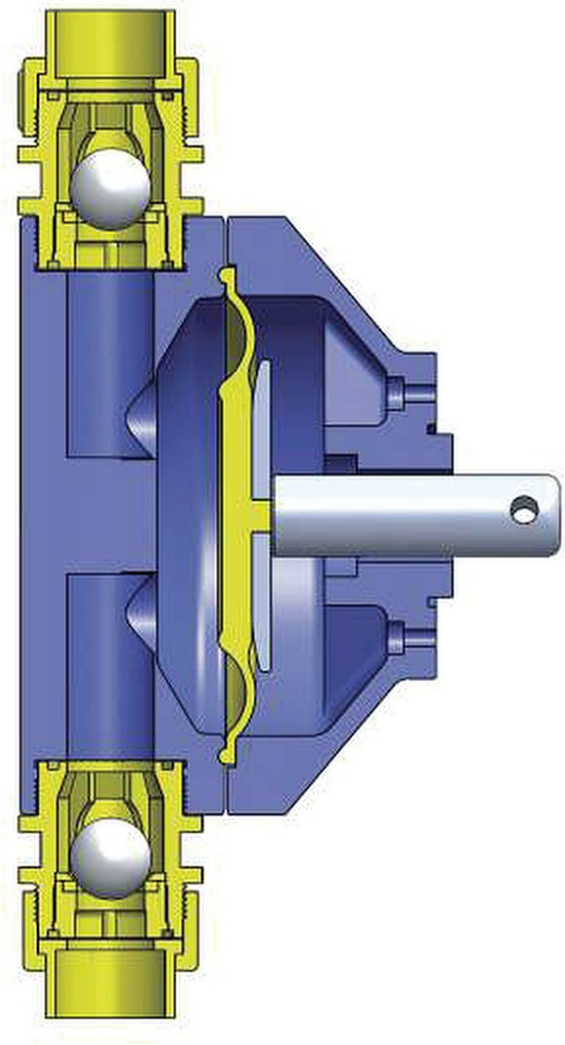
TRUSTWORTHY PUMP · MADE BY AQUARIS

MECHANICAL DIAPHRAGM METERING PUMP

AQUARIS
MADE IN ITALY

DEFINITION

A diaphragm metering pump uses a diaphragm in place of a piston. Driven by the drive mechanism, the diaphragm moves back and forth to complete the suction and discharge strokes of the pump.



FEATURES

- No moving seal, no leakage.
- Can transport high-viscosity media, abrasive pastes and hazardous chemicals.
- Cast aluminum housing: high heat dissipation and light overall weight.
- Multi-layer composite diaphragm: 1st layer – super-tough acid-resistant Teflon film; 2nd layer – EPDM elastic rubber; 3rd layer – 3 mm thick SUS304 supporting core; 4th layer – nylon fiber reinforcement; 5th layer – fully encapsulated in EPDM elastic rubber. This effectively extends the service life of the diaphragm.
- The pressure is generally less than 1.2 MPa.

PUMP HEAD CONFIGURATION ACCORDING TO DIFFERENT APPLICATIONS



PVC



SS304 / 316



PVDF / PTFE

JLM SOLENOID METERING PUMP

AQUARIS MADE IN ITALY

PRODUCT CHARACTERISTICS

- Wear-free solenoid drive with no bearings.
- Adjustment by both stroke length and stroke speed.
- Intelligent LED display shows the stroke speed in percent.
- **Power supply:** 1~ 220–240 V, 50 Hz
- **Material:** PVC; PVDF; SS304; SS316



10–50 L/h	1–3 bar	45 W	1~ 220–240 V
FLOW	PRESSURE	POWER	POWER SUPPLY · 50 Hz

PARAMETER TABLE

Model	Flow (L/h)	Pressure (bar)	Flow (GPH)	Pressure (psi)	SPM	Diaphragm (mm)	Power (W)	Connection
JLM1003	10	3	2.64	43.5	80	41	45	5×8 mm PE hose
JLM2003	20	3	5.28	43.5	140			
JLM3001	30	1	7.93	14.5	180			
JLM5001	50	1	13.21	14.5	140			

MATERIAL CONSTRUCTION

Material	PVC	PVDF	SS304	SS316
Diaphragm	PTFE	PTFE	PTFE	PTFE
Valve body	PVC	PVDF	SS304	SS316
Valve ball	Ceramic	Ceramic	Ceramic	Ceramic
Valve seat	PTFE	PTFE	SS304	SS316
O-ring	Viton	Viton	PTFE	PTFE

JBB MECHANICAL DIAPHRAGM METERING PUMP

AQUARIS MADE IN ITALY

MAIN PARAMETERS

- **Flow range:** 50–521 L/h @ 50 Hz
- **Pressure range:** 5–10 bar
- **Power supply:** JBB50 – JBB155: 3~ 380–415 V / 1~ 220–240 V, 50 Hz as standard. JBB201 – JBB521: 3~ 380–415 V, 50 Hz as standard. Others on request.
- **Material:** PVC; PVDF; SS304; SS316; PTFE

50–521 L/h
FLOW · 50 Hz

5–10 bar
PRESSURE

0.25–0.37 kW
POWER

60 / 112 mm
DIAPHRAGM

PARAMETER TABLE · JBB50 – JBB321



Model	Flow (L/h)	Flow (GPH)	SPM	Pressure (bar)	Pressure (psi)	Diaphragm (mm)	Power (kW)
JBB50/10	50	13.21	100	10	145	60	0.25
JBB75/10	75	19.81		10	145		
JBB101/10	101	26.68	150	10	145		
JBB155/10	155	40.95		10	145		
JBB201/7	201	53.1	120	7	101.5	112	
JBB321/6	321	84.8		6	87.0		

PARAMETER TABLE · JBB421 – JBB521



Model	Flow (L/h)	Flow (GPH)	SPM	Pressure (bar)	Pressure (psi)	Diaphragm (mm)	Power (kW)
JBB421/6	421	111.2	140	6	87.0	112	0.37
JBB521/5	521	137.6		5	72.5		

CONNECTION

Model	PVC	PVDF / PTFE	SS304 / 316
JBB50 – JBB155	DN15 glue union	RC1/2" F	10×16 mm welding union
JBB201 – JBB321	DN15 glue union	RC1/2" F	15×22 mm welding union
JBB421 – JBB521	DN15 glue union	RC1/2" F	15×22 mm welding union